

ABDELRAHMAN GAMAL ZAYED

AI / Machine Learning Intern | Artificial Intelligence Engineering Student

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SUMMARY

Artificial Intelligence Engineering student targeting an AI / Machine Learning Intern role, with hands-on experience in supervised learning, deep learning, computer vision, natural language processing (NLP) basics, data collection, data cleaning, exploratory data analysis (EDA), feature extraction, model evaluation, and data visualization. Built machine learning and image classification projects using Python, Scikit-learn, TensorFlow, Keras, and analytical thinking to support data-driven solutions, intelligent solutions, technical findings, and collaborative AI project development.

EDUCATION

Bachelor of Engineering in Artificial Intelligence Engineering | New Ismailia National University | Oct 2023 - Jul 2028 | Ismailia, Egypt

GPA: 3.65 / 4.00

Related fields: Artificial Intelligence, Machine Learning, Data Science, Computer Science, Deep Learning, Data Structures, Object-Oriented Programming (OOP).

EXPERIENCE

Deep Learning Trainee | Egyptian Engineers Syndicate | Sep 2025 | Online

- Completed a 2-week online training covering neural networks, validation metrics, overfitting control, transfer learning concepts, and practical deep learning applications.
- Applied motivation, analytical thinking, and technical documentation to summarize model training logic and experiment results.

Machine Learning Intern | National Telecommunication Institute (NTI) | Jul 2025 | Remote

- Built a diabetes prediction workflow using structured health data, data cleaning, preprocessing, feature engineering, model training, and imbalance-aware evaluation.
- Compared 4 supervised learning algorithms - Logistic Regression, Random Forest, Support Vector Machine (SVM), and Decision Tree - using accuracy, precision, recall, F1 score, and confusion matrix validation metrics.

AI & Machine Learning Trainee | SprintUp Program by Sprints | Apr 2025 - Jun 2025 | Remote

- Developed an end-to-end supervised machine learning workflow using Python, Scikit-learn, exploratory data analysis, data visualization, model building, and result interpretation.
- Transformed raw data into model-ready features and documented technical findings for collaborative project development.

RESEARCH EXPERIENCE

Research Paper - Hybrid Metaheuristic Model for High-Fidelity Solar Power Forecasting | Under Review

- Co-authored a research paper integrating Extreme Learning Machine (ELM) with Improved Harris Hawks Optimization (IHO) for solar power time-series forecasting.
- Evaluated forecasting performance against CatBoost, XGBoost, Transformer, and standalone ELM using R2, Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

PROJECTS

Diabetes Prediction Classification Model | Machine Learning | Healthcare Data

- Built and evaluated classification models for diabetes risk prediction using data preprocessing, feature selection concepts, hyperparameter tuning basics, cross-validation basics, and model comparison.
- Reported performance using 5 evaluation metrics: accuracy, precision, recall, F1 score, and confusion matrix; focused on recall and class-level error analysis.

Facial Emotion Recognition Model | Deep Learning | Computer Vision

- Developed a 7-class image classification workflow for facial emotion recognition using TensorFlow/Keras, convolutional neural network (CNN) concepts, data augmentation, validation monitoring, and transfer learning concepts.
- Used class-level error analysis, confusion matrices, and overfitting checks to improve model evaluation quality; applied OpenCV basics for computer vision preprocessing concepts.

NLP Learning Mini-Tasks | Natural Language Processing | Text Classification

- Practiced text preprocessing, feature extraction concepts, and basic text classification workflows as part of AI/ML internship preparation and self-learning.

SKILLS

Programming & Tools: Python, C, C++, Jupyter Notebook, Git, GitHub, PyCharm, technical documentation.

Machine Learning & Data Science: Machine Learning, supervised learning, classification, regression basics, data collection, data cleaning, exploratory data analysis (EDA), data visualization, feature engineering, feature selection, hyperparameter tuning, cross-validation, model evaluation, validation metrics.

Libraries & Models: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, TensorFlow, Keras, PyTorch basics, OpenCV basics, Logistic Regression, Random Forest, Support Vector Machine (SVM), Decision Tree, XGBoost basics.

AI Areas: Deep Learning, Neural Networks, Computer Vision, image classification, Natural Language Processing (NLP), text preprocessing, feature extraction, text classification, model deployment basics, intelligent solutions, data-driven solutions.

Soft Skills: Problem-solving, analytical thinking, collaboration, teamwork, motivation, time management, fast learning, communication. Languages: Arabic native; English intermediate.

LEADERSHIP & VOLUNTEERING

AI Head | Microsoft Students Club - Suez Canal University (SCU)

- Lead AI-focused student activities and support peer learning in machine learning, deep learning, NLP, and computer vision topics.

Volunteer | Egyptian Youth Entity (EYE)

- Contribute to youth development and community initiatives, strengthening collaboration, communication, and leadership skills.